L12.1: Higher order partial derivatives and the Hessian matrix

Transcript Language

English

Hello, and welcome to the Maths 2 component of the online BSc program on Data Science

and Programming. This video is about higher order partial derivatives and the Hessian

matrix. So, in the earlier video, we studied the notion of critical points for multivariable

functions. So, what that meant was that you take the function f is a scalar valued multivariable

function, and then you find its gradient and then you set it to 0.

And you compute all those points for which those equations are satisfied, which means

that the gradient at that point is 0 or the other thing that can happen is that one or

more partial derivatives are undefined. So, you collect together all those points and

those are called the critical points of F.

And the reason we were interested in studying these was that the local extrema, meaning

local maxima or local minima are all critical points. So, they satisfy that the gradient

is 0, or maybe some partial is not defined. So, they must satisfy one of these. And the

intuition there was that the tangent plane is parallel to the x-y plane or in general

x1 x2 xn hyperplane.

In one variable calculus, we have studied that there is a second order test, meaning

a test using the second derivative, and that allows us to classify which of these critical

points are local maxima, local minima or saddle points, and sometimes it can fail also. So, we would like an analogue of such a test for the multivariable case. And in order to do

that, we will first study the notion of higher order partial derivatives and the Hessian

matrix.

So, let us recall what are partial derivatives first, suppose f of x1 x2 1 is a scalar valued

multivariable function defined on a domain D in Rn. The partial derivative of f with

respect to the variable xi is the function denoted by f subscript xi, or del f del xi.

And it is defined by taking the limit as h tends to 0, f of x tilde plus h times ei,

what is ei, ei is the ith unit vector in Rn minus f of x tilde divided by h.

So, this is the definition of the partial derivative function of f with respect to the

ith variable xi. And this domain, it need not always, this limit may not always be,

it need not always exist. And so the domain of this function, the partial derivative is

all those points within D where this limit does exist.

And what does the partial derivative do, the partial derivative at a point measures the

rate of change of the function in the direction of the standard versus vector ei at the point

a tilde. So, or equivalently with respect to the variable xi, meaning it treats the

function as a function of only xi.

So, you restrict to that line parallel to the xi axis passing through a tilde and restrict

your function to that line and then ask what is the, how is the function changing? What

is the rate of change of that function? That is exactly what the ith partial derivative

does.

So, now the partial derivative is a function. And so, you can say what is the partial derivative

of that function, the partial derivative is a scalar valued multivariable function, and

we can very well ask what is the partial derivative of that function and that is exactly what

are second order partial derivatives for f x, y.

So, if f of x, y is a function defined in a domain D in Rn, then the second order partial

derivatives of f are the partial derivatives of the partial derivatives. So, you take partial

derivatives and then you again take partial derivatives. So, just for the sake of simplicity,

the notations are either f subscript xx or del squared f del x squared. So, note this

strange thing that the notation here is del squared f del x squared and not del squared

f del squared x. So, this is for something that will come ahead.

The other possibility is that you have fyy which is also del squared f del y square and

the third possibility is that you have fxy, which is the same as del squared f or del

y del x or and the fourth possibilities fyx, which is del squared f del x del y. So, what

do these mean? So, fxx means, you consider the partial derivative f with respect to x,

of f with respect to x and then take the partial derivative of that function again with respect

to x.

So, f subscript x again subscript x or the other way of writing this del del x of the

function del f del x. So, del f del x is the function the partial with respect to x and

then you take the partial again with respect to x. Similarly, if you have fyy, what that

means is you take partial with respect to y and then again take partial with respect

to y equivalently you could write it as del del y of del f del y. What is fxy?

So, this is where you first take the partial with respect to x and then take the partial

with respect to y of that function, which is the same as saying you take del del y of

del f del x and then here this is fy subscript x and this is del del x of del f del y. So,

now, just notice one small thing, this is a fairly obvious but sometimes students make

mistakes with this.

In the notation when you do fxy so, here the order is first differentiate with respect

to x, then take partial with respect to y, but when you write it like this, it gets kind of flipped over, so you have to remember that this means first take partial with respect

to x, and then take partial with respect to y. So, because of the notation on the right

is sort of saying you take del del of something and that is why it is written like this.

Whereas, on the left, it is x is coming on the right, the variables are coming on the

right, that is why you write it like this. So, you have to just remember that when you

go between these 2, whatever is written here, gets it in the opposite order towards the

left, so the subscript notation gets written towards the right, the del del notation gets

written towards the left, this is what you should remember, because this is a common

source of errors.

Now, it would be of course, convenient if we could flip these as we want. Like, if we

have fxy and fyx, then we can interchange them. That would be very convenient. And indeed,

we will soon see a theorem that says that if your function is nice, meaning under some hypothesis, indeed, that does happen.

Let us start with some examples. So, let us compute all the second order partial derivatives

of these functions. So first, what are the first order partial derivatives, so del f

del x. So, we have actually computed these before is 1, del f del y is 1. And so del

squared del x squared is the partial of the function del f del x, which is 1, so the partial

of that is 0, because it is a constant, then del squared f del y squared is similarly 0,

in fact all these functions are 0; del squared f del x del y is also 0 and del squared f

del y del x is also 0.

And indeed do notice that here, these two are equal. So, at least for easy functions,

it may happen that these two are often equal. So, it does not matter in which order you

do your partial differentiation, the answer will be the same. Let us see if that holds

for this as well. So, you have f of x, y is sin of x, y, we have again computed the gradient here, so we know what the partial derivatives look like.

So, this is y times cosine of x y, and x times cosine of x y. Now, let us look at the higher

order partial derivatives. So, if you have del squared f del x squared, that means you

have to differentiate y times cosine x, y with respect to x. Now y is a constant, so

it will come out so you are just differentiating cosine of x y. So, this is y times, y times

minus sin of x y times y. So, this is minus y squared sin of x y.

Now, by symmetry or by doing it, you can see that del squared f del y squared is also minus

x squared times sin x y. So, I will suggest you check this. So, this leaves us with the

mixed partial derivatives. So, del x del y and del squared f del y del x. So, for del

squared f del x del y, we first differentiate with respect to y so that gives us x cosine

x, y and then differentiate with respect to x. So, this is your product rule.

So, let us differentiate x first, that is 1, so 1 times cosine of x y and then hold x and differentiate cosine of x y. So, that gives us y times, minus y times sin of x y.

So, what this gives us is cosine of x y minus x y times sin of x y. So, that is what it

gives us. And then by symmetry or by computing it, maybe we can compute it. So, this is again

1 times cosine of x y. So, I am now differentiating y times cosine x with respect to y. So, 1

times cosine of x y plus y times minus x, sin of x y, which gives us cosine of x y minus

x times, minus x y times sin of x y.

And again, you can see that these two are indeed equal. So, it seems that the mixed

partials are indeed often equal. Maybe I should have mentioned in the previous slide that

these are called the mixed partials, the mixed partial derivatives. So, as I was saying,

the mixed partial derivatives, at least in these examples seem to be equal.

And indeed, we have a very nice theorem called Clairaut s theorem about the mixed partials.

So, it says the following, so suppose f of x, y is a function defined on a domain D in

R2 containing a point A and an open ball around it. If the second order mixed partial derivatives,

fxy and fyx are continuous in an open ball around a, then they are equal at a.

So, what do we need to check that these mixed partial derivatives are continuous in an open

ball around a. Now, in our previous examples, they were all very nice functions. So, continuity

was not at all a problem. And from there, we could as a result, apply Clairaut s theorem.

And we saw that they are equal although we actually computed them, but we could have

directly concluded it from Clairaut s theorem.

So, let us see an example where we will try to compute all these and see that the hypothesis

is indeed necessary. So, let us compute what are the partial derivatives. So, first of

all, what is the left del x and del f del y at 0, 0. So, del f del x at 0, 0 is limit

h tends to 0, f of 0 plus h comma 0 minus f of 0 comma 0 divided by h. So, f of 0 plus

h, that is f of h comma 0, that is again 0 minus 0 by h, this is 0 and so this is going to be 0 by the same argument, the partial for y is also 0.

So, we know that these two are 0. Of course, we have to compute now the partial derivatives

at the other points as well. So, now let us do that. So, what is del f del x at x y where

x y is not equal to 0, 0. So, x, y is not equal to 0, 0, then this is a rational function

and the denominator is non-zero so I can apply my u by v rule. So, if we do that, let us

see what we get.

So, this is x squared plus y squared times the derivative of the numerator. So, that

gives us y time x squared minus y squared plus x y times 2x minus x y times x squared

minus y squared times 2x divided by x squared plus y squared. So, now if we simplify this,

and I will not spend time on, on doing this, since it is a direct simplification, what

we are going to get is something like minus y to the power 5 plus x to the 4 times y,

plus 4x squared y cubed divided by x squared plus y squared squared. And this is if of course, x, y is not equal to 0.

And you can do the same thing for del f del y. And if we do that, what you will get is,

again, you use a u by v rule and go through the expressions, you will get exactly the

same thing. Except that now the x term the highest x term comes with a positive sign

and the lower ones come with a negative sign. So, x cube y squared divided by x squared

plus y squared squared.

So, it is kind of symmetric, except that there is this gap between the plus and the minus

and that gap is exactly what is going to cause the second order partials to not work out

properly. So, so this is what is. And now let us compute what is the second order partial,

mixed partials for at 0, 0. So, at 0, 0, I want to compute what is del squared of del

x del y 0, 0.

So, what does this mean this means I have to compute limit h tends to 0 the partial of y at, so this with respect to h, so h comma 0 minus del f del x at 0, 0 divided by h.

So, what is the partial derivative, that h comma 0 this is so limit h tends to 0, h to

the power 5 minus everything else is 0 because the y coordinate is 0, and then divided by

the numerator and then divided by x squared plus 0 squared squared, which is h to the

power 4.

So, we will simplify this in a minute, what is the partial of x at 0, 0 that we have computed

that is 0 and then divided by h. So if we do this, this is h to h minus 0 by h limit

h tends to 0 which is 1. Now, let us compute delta squared f del y del x at 0, 0. So, this

is limit h tends to 0 the del f del x at 0 comma h minus del f del x at 0 comma 0 I seen

the previous one, I have made a typo this should have been del f del y but it is the

same, the answer does not change as such divided by h and now we can again do the same thing.

So, what is del f del x at 0 comma h. So, we have found out here that it is minus h to the power 5 and all the other terms do not contribute because there is a x in there

and the x we have to substitute is 0. So, minus h to the power 5 and then the denominator

is 0 squared plus x squared squared minus 0 divided by h and limit as h tends to 0 and

you can see now this is minus 1 and so these do not match. So, this do not match.

So, what was the problem? The problem was that these mixed partials are actually not

continuous. So, for that, you will have to evaluate what they are and check what happens.

So, the hypothesis in Clairaut s theorem is important, and without that, it may not be

happen that the mixed partials are indeed equal. So, for whatever is next we will assume

in that the hypothesis of Clairaut's theorem holds, because we want that to hold in order

to get whatever the second order partials are useful for fine.

So, we have done second order partial derivatives for fxy. Now, let us talk about second partials

for f x1 x2 xn. So, if f x1 x2 xn is a function defined on the domain D in Rn, then the second order partial derivatives are defined analogously as a partial derivatives of the partial derivatives.

So, here there are for the n is 2 case there were 4 second order partial derivatives. So,

for the general case there are going to be n squared second order partial derivatives.

So, they are given by f xi xi or del squared f xi xi. So, what this means is you take f

with respect to xi and again with respect to xi or the same thing del del xi of del

f del xi. This is when i is equal to j if i is not equal to j or meaning this is a general

situation, then this is f xi and then you take the partial of xj of that which is the

same as saying you take del del xj of del f del xi.

So, again, important point is to notice here how here it is written on the right, and so

it goes, the order is from left to right, here it is written towards the left and so

the order is from right to left. That is the only thing you have to recall, remember, as

far as notation is concerned. And the other difference meaning why did I write it differently?

See, here the notations are same f xi xi, here xixj. So, it does not matter if i and

j are different or i and j are the same, but when you write it in terms of this del notation,

the del xi del xi becomes del xi squared. This is just convention, there is no reason

for this notation other than convention.

Fine, let us do this example. If x y is xy plus yz plus zx, let us first compute the

partial so del f del x is y plus z. Again we have done this example before meaning the

partials first order partials before this is x plus z or z plus x and del f del z is

x plus y. And so if I want the second order partials, I differentiate these. So, del squared

del x squared is 0 del squared del x del y del x is. So, you take the function y plus

z and differentiate with respect to y. So, this is 1 del squared del f del z del x is

1.

And I think you can see the general pattern emerging so I will write it as del squared del y squared is 0 del squared f del z squared is 0 and then del squared f del z del y this

is the partial of the function del f del y which is x plus z. So, partial of x plus z

with respect to z that is 1 and again if we know that the partials are, the second order

partials are nice then the mixed partials are actually the order does not matter.

So, from Clairaut's theorem or an equivalent version of the theorem in n dimensions, you

can check that del squared f del x del y is also equal to 1 which is also del squared

f del y del z which is 1 and then del squared f del maybe I want a del x here and then del

y del z all these are 1. So, you can check this directly from the expressions for del

f the first order partials or you can use the fact that Clairaut's theorem applies.

So, I will make that as a comment in this slide. So, before which let us define the

higher order partial derivatives. So, you can keep repeating this process. So, you have the second order partials you can take the third order partials, fourth order partials.

So, for one variable functions, you can differentiate as many times as you want provided the derivative

exists. So, the same thing can be done for partial derivative, you can take partial derivatives

as many times as you want. And that is exactly what higher order partial derivatives do.

So, if f x1 x2 xn is a function defined in a domain D in Rn, then the higher order partial

derivatives of f are defined analogously by taking successive partial derivatives. So,

the general notation so, if you take the kth order partial derivative, the notations are

f xi 1, xi 2, xi k which means you take the partial with respect to xi 1, then you take

the partial with respect to xi 2, then you take the partial with respect to xi 3 and

all the way up to xi k, this is what it means.

And you can write the same thing except remember our usual thing about the shift in the order.

So, this is exactly the same as taking del del xi of xi k of del del xi k minus 1 of del del xi k minus 2 of all the way up to del f del xi 1 and then plenty of brackets,

however. So, I hope the idea is clear, it is exactly the same thing. Of course, once

again one should point out that there is no guarantee that this higher order partial derivative

exists the same way as there is no guarantee that the first order partial derivative exists.

So, there will be some points where it exists some points it may not exist and so, the domain

of this function is whichever points it exists and an appropriately modified statement of

Clairaut's theorem holds. What that means is, if suppose you take the kth order partial

derivatives, then if all the, if this guy is continuous and so is in the neighborhood

and so is any, you take this denominator in some other order, if that kth order partial

is also continuous in some neighborhood, then they are going to match at the point, at any

point in that neighborhood.

So, in general what you can say is under suitable hypothesis, you can shift the orders without any problem. So, if I do xi 1, xi 2, xi k that is the same as doing it in some other

order xij 1, xij 2, xij k. So, some confusing combinatory it maybe, but all I am saying

is you change the order of 1 1 through k instead of x1 x2 x3, you could have x2 x3 x1 and that

is allowed under suitable hypothesis.

Finally, let us talk about the Hessian matrix which is really what all this is going to

be used for. So, the Hessian matrix is the following if you have a function of n variables

defined on domain D in Rn, then the Hessian matrix of f is defined as this complicated

looking n by n matrix. So, remember that you had n squared second order partial derivatives,

so, you place them in a matrix.

So, the first row is you take all your, you take your gradient vector and then you take

the partial with respect to x1 of each component of that gradient vector. So, del squared f

del x1 squared del squared f del x1 del x2 del squared f del x1 xj. So, this is the jth column del squared f del x1 del xn that is the last column. And then for the second row

you take del del x2 of the gradient vector. So, del del f del squared f del x2 del x1,

del squared f del x2 del x2, then del squared f del x2 del xj, del squared f del x2 del

xn and then in general this is ith row.

So, the important part is you have to remember what happens for the ijth entry. That is how

you describe any matrix by saying what happens to the ijth entry. So, the ijth entry is given

by taking del squared f del xi del xj. This is what if you remember. So, that is what

is happening. So, this is the Hessian matrix and this is somehow going to come in, this

is somehow going to be useful when we talk about the second derivative test whatever,

so the analog of the second derivative test when we classify critical points.

Fine, let us do a couple of examples. So, for this example, x plus y, we saw that the

second order partials are actually all 0. So, the Hessian here is 0 0 0 0. So, this was a relatively complicated Hessian, sin x y. So, it is of, it is a 2 by 2 because

you have two variables. And the terms here are. So, this is minus y squared sin x y minus

x squared sin x y and then this was a cosine of x y minus x y times sin of x y then this

was cosine of x y minus x y times sin of x y. So, these terms typically match. So, often

for the Hessian matrix, these terms will match.

And then let us do this one. This one was again rather easy matrix. Now, this is a 3

by 3 matrix, why 3, because there are 3 variables. So, what was this matrix? This was 0 1 1;

1 0 1 and 1 1 0, the kind of matrices we found in linear algebra were somewhat useful. And

indeed, we will have some use of linear algebra when we study this hessian matrix in the context

of the classification of critical points. Thank you.